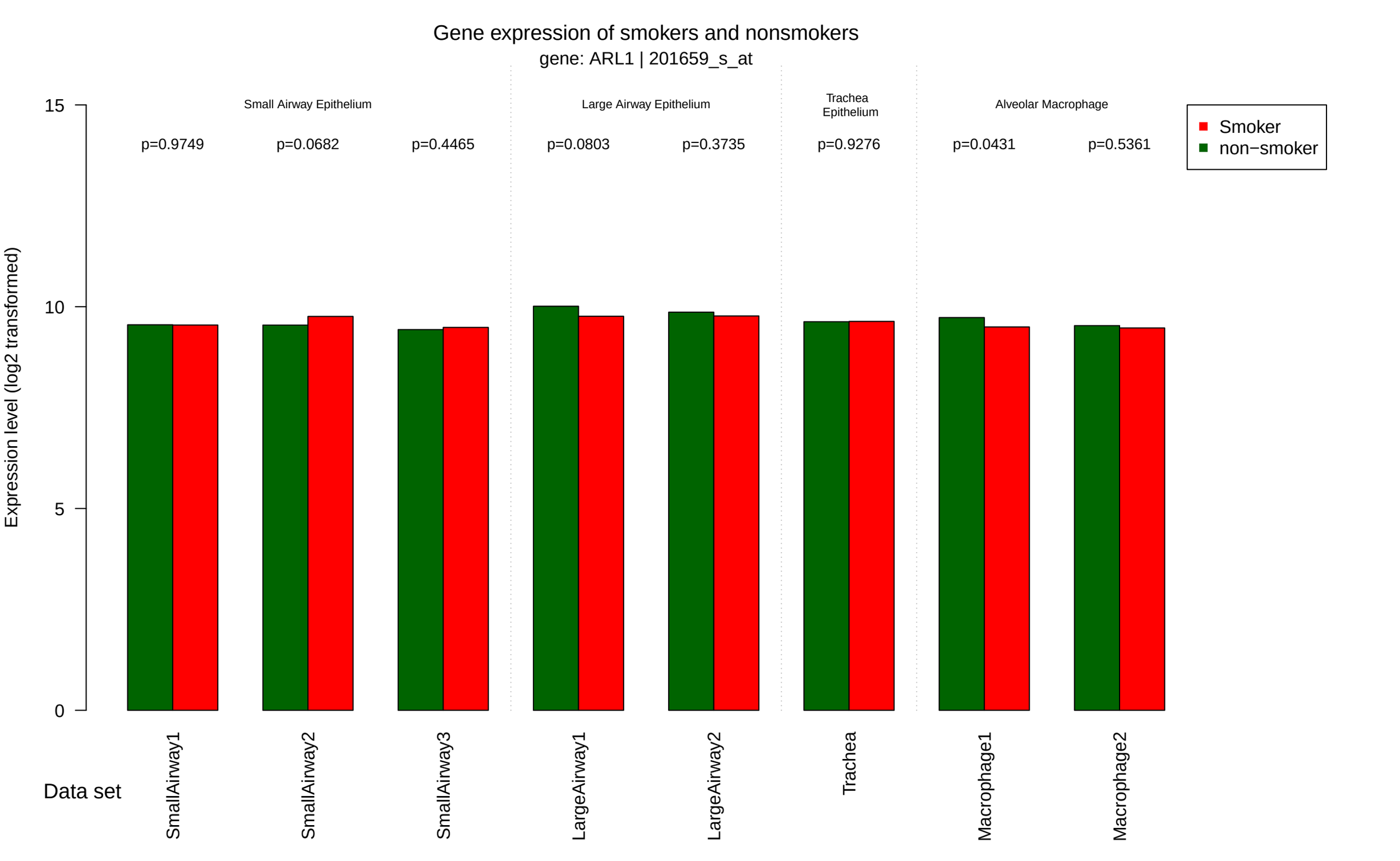
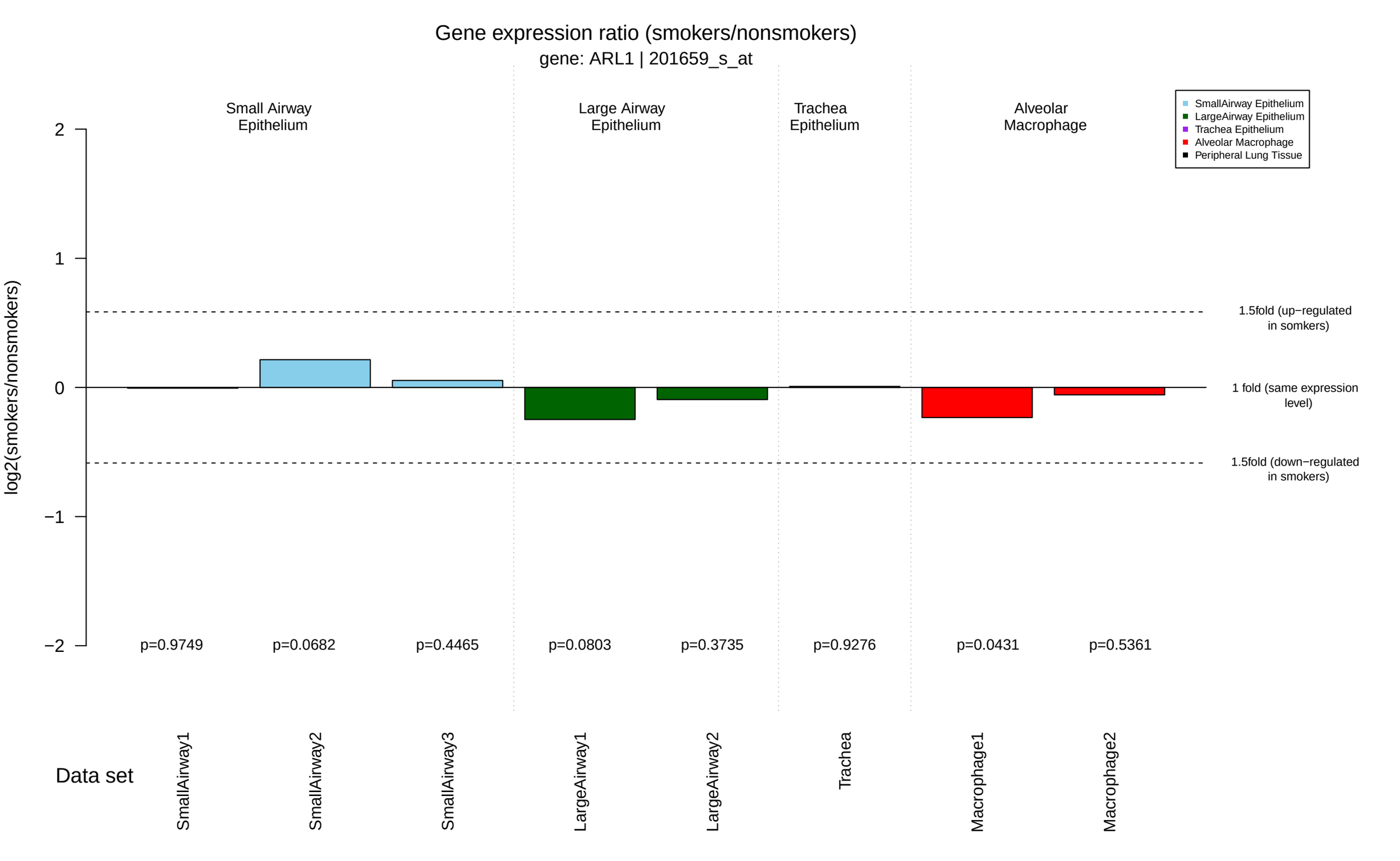
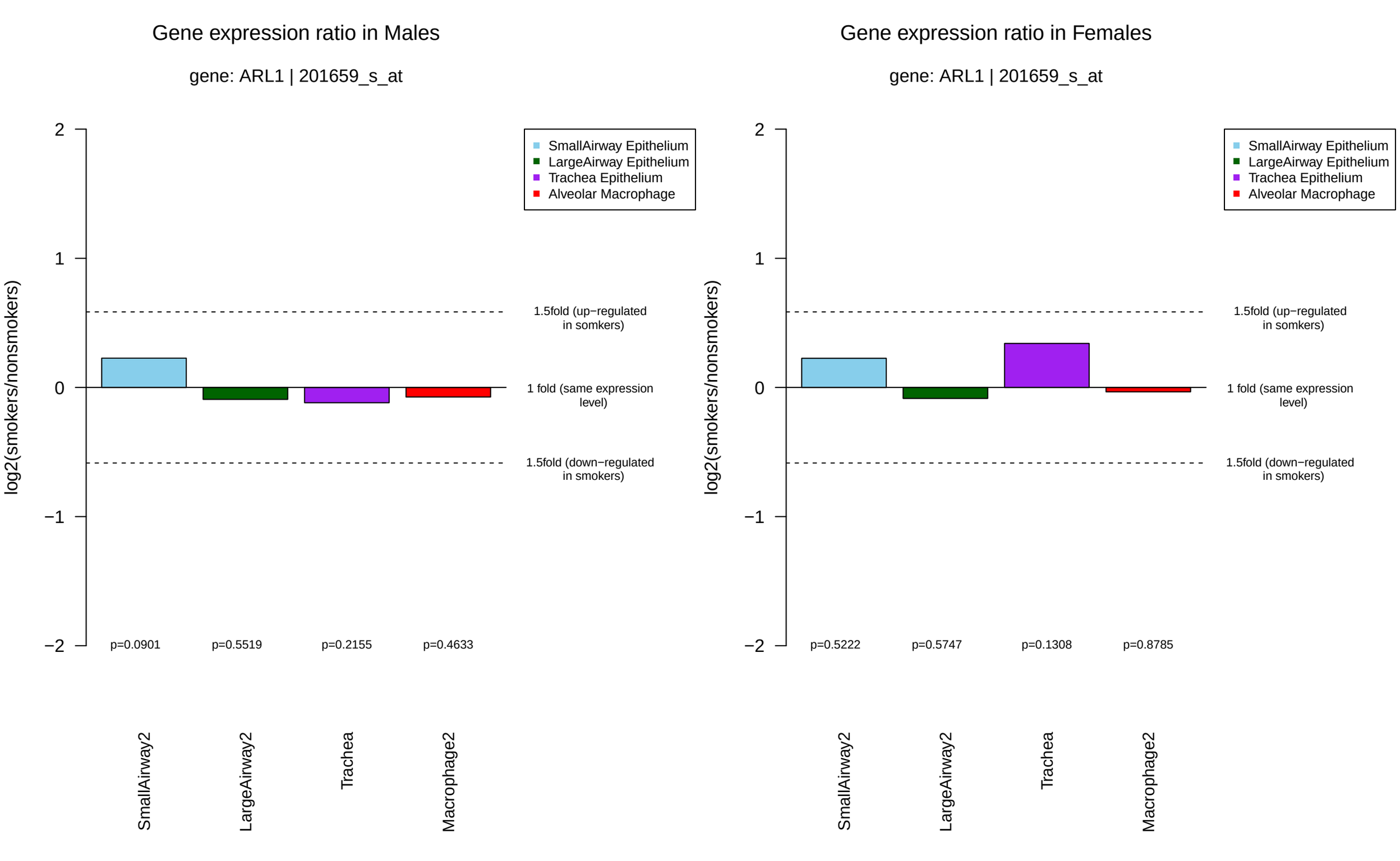
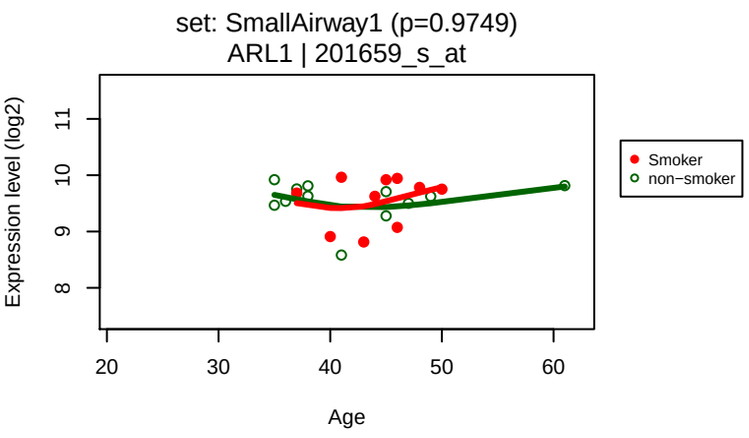
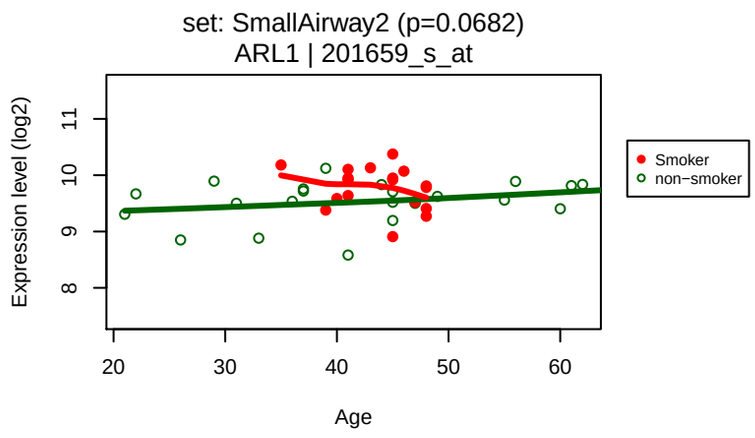
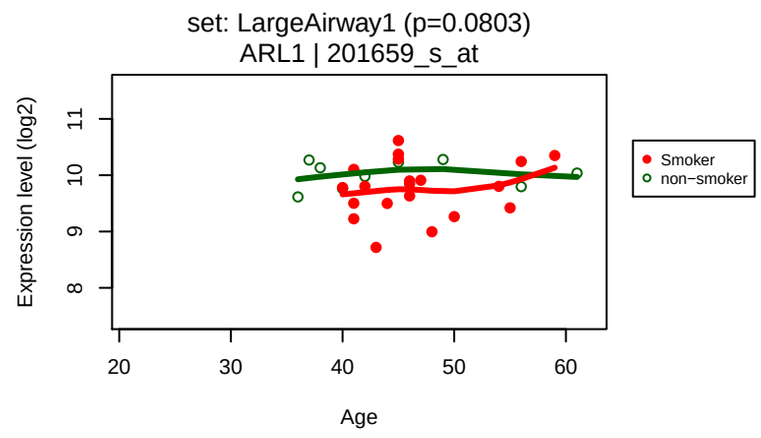

Gene expression of smokers and nonsmokers
gene: ARL1 | 201659_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9749
p=0.0682
p=0.4465
p=0.0803
p=0.3735
p=0.9276
p=0.0431
p=0.5361
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

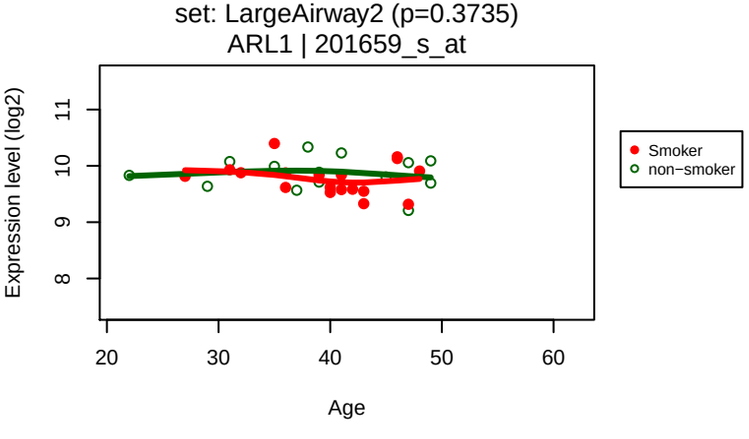

Gene expression ratio (smokers/nonsmokers)
gene: ARL1 | 201659_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9749
p=0.0682
p=0.4465
p=0.0803
p=0.3735
p=0.9276
p=0.0431
p=0.5361
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

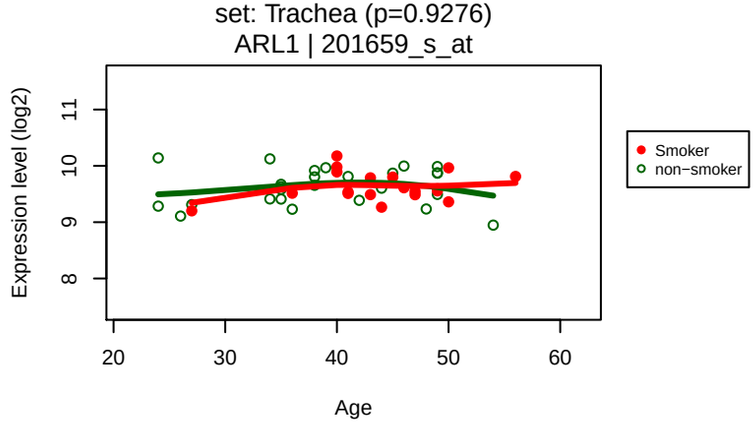

Gene expression ratio in Males
gene: ARL1 | 201659_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0901
p=0.5519
p=0.2155
p=0.4633
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL1 | 201659_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5222
p=0.5747
p=0.1308
p=0.8785
SmallAirway2
LargeAirway2
Trachea
Macrophage2

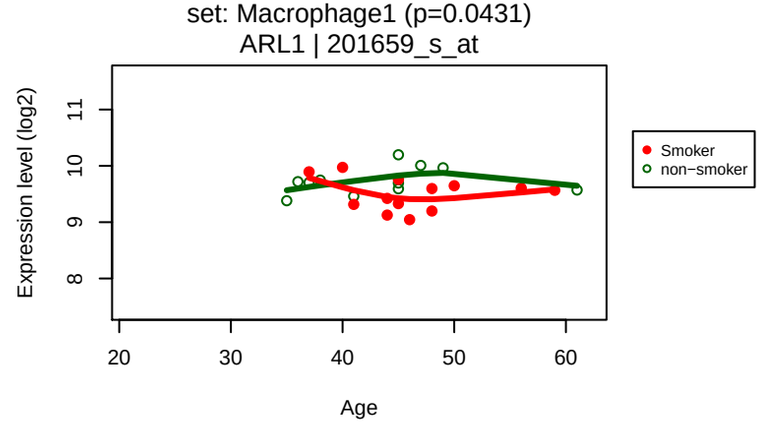

set: SmallAirway1 (p=0.9749)
ARL1 | 201659_s_at
Expression level (log2)
Age
Smoker
non−smoker

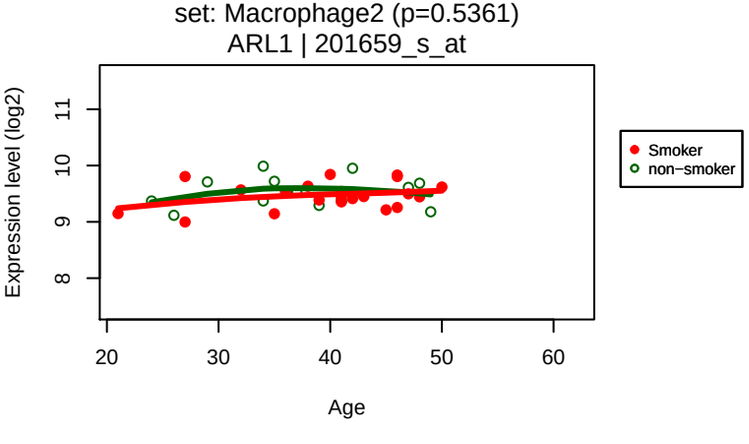

set: SmallAirway2 (p=0.0682)
ARL1 | 201659_s_at
Expression level (log2)
Age
Smoker
non−smoker

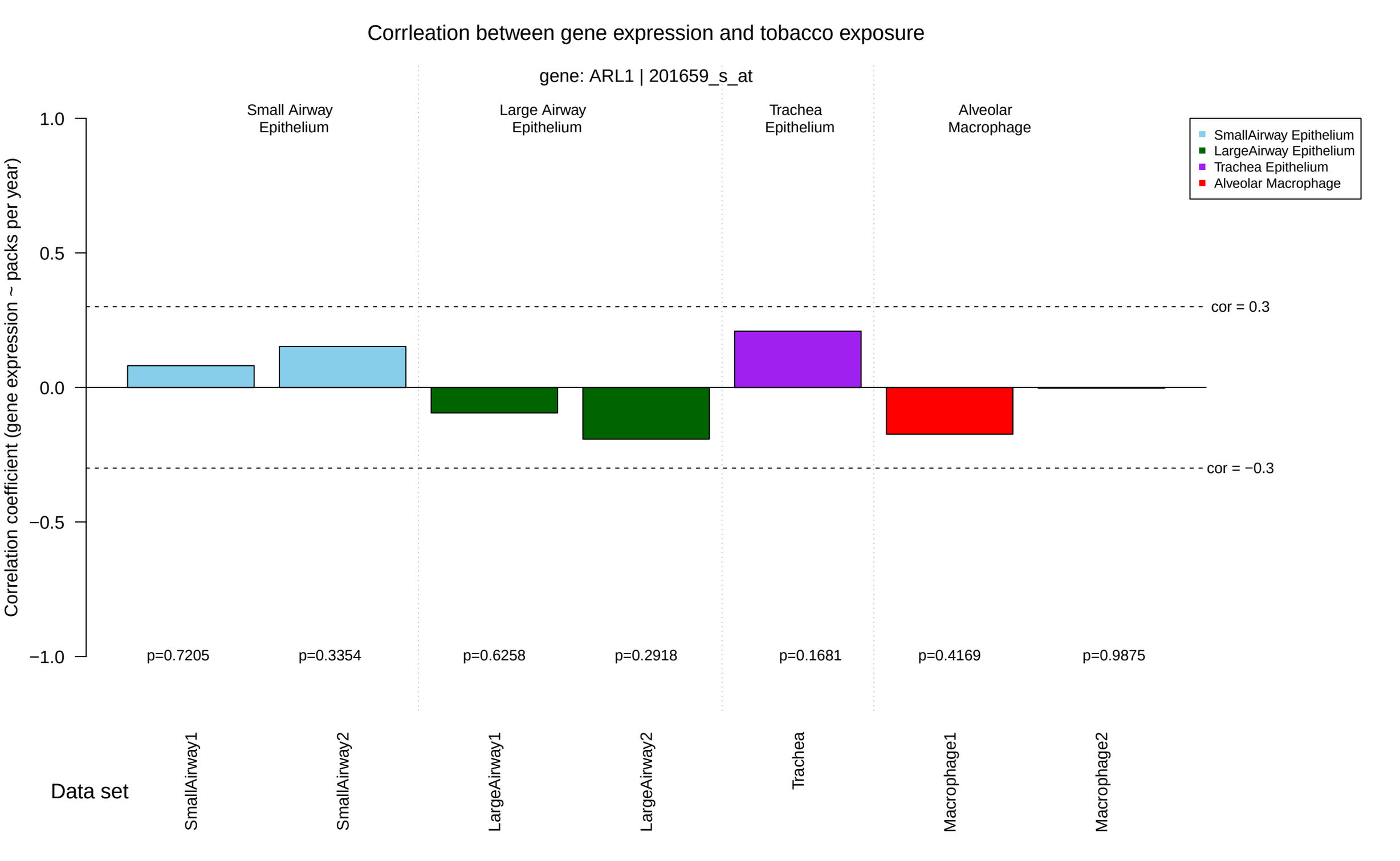

set: LargeAirway1 (p=0.0803)
ARL1 | 201659_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3735)
ARL1 | 201659_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9276)
ARL1 | 201659_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0431)
ARL1 | 201659_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5361)
ARL1 | 201659_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL1 | 201659_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7205
p=0.3354
p=0.6258
p=0.2918
p=0.1681
p=0.4169
p=0.9875
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2